

Matteo Paolini

Mechanical Engineering M.Sc. student | Mechatronics & Robotics

Brussels / Italy · Open to relocation

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RESEARCH EXPERIENCE

Research Intern - Royal Military Academy, Brussels | 2026 - Present

DREAM project · Work in progress. Research on Physical AI for robotic manipulation, with a focus on reinforcement learning, hierarchical RL and sim-to-real transfer. Developing reproducible MuJoCo scenes, physics diagnostics and Gymnasium-based learning experiments; exploring counterfactual reasoning for physical interactions in cluttered scenes.

EDUCATION

M.Sc. Mechanical Engineering - Mechatronics | UNIVPM | 2024 - Present

Erasmus coursework at Université Libre de Bruxelles (ULB) and Vrije Universiteit Brussel (VUB), 2025-2026. Robotics, control, industrial automation and data science.

B.Sc. Mechanical Engineering | UNIVPM | 2021 - 2024

Thesis: motion planning for redundant robots (TIAGo). Implemented and compared RRT and RRT* for collision-free pick-and-place trajectories.

SELECTED ENGINEERING PROJECTS

Dual-arm Object Handover | Individual project | 2025-2026

Developed the complete Drake/OMPL simulation pipeline for two Franka Panda arms: workspace analysis, inverse kinematics, motion planning, grasp ownership and task coordination. Demonstrated a 13-state pick-handover-place sequence; one nominal simulated run completed in approximately 20 seconds with 9.1 mm placement error.

Tentacle Robotic Gripper | Team project | 2025

My contribution: hyperelastic FEM of the pneumatic silicone actuator, casting-mold CAD and 3D-printed tooling. Integrated by the team into a soft gripper and two-degree-of-freedom arm; public build guide and prototype video.

Apple Watch Case - Manufacturing Study | Academic project | 2025

Reconstructed case geometry and prepared manufacturing drawings. Developed a proposed high-volume production workflow, with machining calculations, tool-life estimates and sheet nesting. Material and geometry assumptions support an academic process-design study.

vacUUUm - Cordless Vacuum Design | Team project | 2025

Connected user research and QFD to functional decomposition and product architecture. Worked on Siemens NX CAD, assembly layouts and technical drawings for a cordless vacuum and charging station.

CubeSat 1U Structure | Academic project | 2025

Studied a hybrid chassis with composite panels and aluminium rails using ANSYS/ACP. Defined a symmetric 16-ply, 2 mm laminate and investigated structural design and multi-material integration.

SKILLS

Robotics & software: Python, MuJoCo, Gymnasium, Drake, OMPL, MATLAB/Simulink.

Engineering: Siemens NX, ANSYS Workbench/ACP, FEM, hyperelastic modelling, mold design, technical drawings, FDM 3D printing.

Materials: metallographic sample preparation, microscopy and Vickers hardness testing.

Languages: Italian (native), English (C1).